ModelIt!

Student Activity Pack

Where Does a Tree's Mass Come From?

A 1-Ton Tree Got Most of Its Weight From Air

Grade 5 | NGSS 5-LS1-1 | Matter and Energy

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 A giant tree weighs 1 million pounds. Where do you think all that mass came from?

2 What do plants need to grow? List everything you can think of.

3 A scientist grew a tree in a pot for 5 years. The tree gained 164 lbs. The soil lost 2 ounces. What does this prove?

4 What is photosynthesis and why do plants need it?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

CO2 in Air | Sunlight | Photosynthesis Rate | Tree Biomass

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Normal Growth

Normal CO_2 and sunlight levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: CO_2 Removed

Lock CO_2 to 0%

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Maximum Growth

Both CO_2 and Sunlight at HIGH

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Water — H_2O is also a raw material for photosynthesis

Temperature — Affects photosynthesis rate - too hot or cold slows it

Chlorophyll — The green pigment that captures light energy

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Where does a tree get the materials it needs to grow? Use evidence from your simulation to support your answers.

1 If you weigh a seed, plant it, and weigh the full tree, where did the extra mass come from?

2 Why is deforestation a climate problem? (Think about stored carbon.)

3 If CO2 in atmosphere is rising, does that mean plants should grow faster?

4 Why can't plants grow in completely dark environments?

5 What does the photosynthesis equation tell us about inputs and outputs?

8 STEM Challenge: Design a Carbon Capture Forest

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YOUR ENGINEERING MISSION

Design a reforestation project to capture the most CO_2 possible.

THE SCENARIO

Your city wants to offset carbon emissions by planting trees. Design the most effective carbon capture forest.

Engineering Design Thinking

1. WHAT TREE SPECIES WILL YOU CHOOSE AND WHY?
2. HOW WILL YOU ARRANGE THE TREES FOR MAXIMUM SUNLIGHT?
3. HOW WILL YOU ENSURE TREES SURVIVE LONG ENOUGH TO GROW LARGE?
4. HOW WILL YOU CALCULATE TOTAL CARBON CAPTURED?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Plant Biologists and Agricultural Scientists optimize photosynthesis to feed 8 billion people. They earn $65,000-$105,000/year.